Breaking down corn insect protection technologies

PIONEER®

ABOVE-GROUND INSECT PROTECTION

		Above-ground insect protection event (protein)	Herbicide tolerance (HT) (event/gene)	Refuge requirement - Corn belt unless otherwise designated
Options in the Pioneer® brand portfolio	**Optimum® AcreMax® & Optimum® Intrasect®** PRODUCTS	TC1507 (Cry1F), MON810 (Cry1Ab)	glyphosate (NK603), glufosinate (pat)	**Optimum AcreMax products: INTEGRATED*** 5% glyphosate, glufosinate HT
				Optimum Intrasect products: STRUCTURED 5% corn belt / 20% EPA cotton counties
	Optimum® AcreMax® Leptra® & Optimum® Leptra® PRODUCTS	TC1507 (Cry1F), MON810 (Cry1Ab), MIR162 (Vip3Aa)	glyphosate (NK603), glufosinate (pat)	**Optimum AcreMax Leptra products: INTEGRATED*** 5% glyphosate, glufosinate HT
				Optimum Leptra products: STRUCTURED 5% corn belt / 20% EPA cotton counties
	PowerCore® Enlist® corn PRODUCTS	TC1507 (Cry1F), MON89034 (Cry1A.105 + Cry2Ab)	2,4-D choline + FOP (DAS40278), glyphosate (NK603), glufosinate (pat)	**PowerCore Enlist Refuge Advanced corn: INTEGRATED** 5% 2,4-D choline, glypohsate, glufosinate, FOP HT
				PowerCore Enlist corn: STRUCTURED 5% corn belt / 20% EPA cotton counties
	PowerCore® Ultra Enlist® corn PRODUCTS	TC1507 (Cry1F), MON89034 (Cry1A.105 + Cry2Ab), MIR162 (Vip3Aa)	2,4-D choline + FOP (DAS40278), glyphosate (NK603), glufosinate (pat)	**PowerCore Ultra Enlist Refuge Advanced corn: INTEGRATED** 5% 2,4-D choline, glyphosate, glufosinate, FOP HT
				STRUCTURED 5% corn belt / 20% EPA cotton counties
Competitive products	**VT Double PRO® corn**	MON89034 (Cry1A.105 + Cry2Ab)	glyphosate (NK603)	**VT Double PRO RIB Complete Corn Blend: INTEGRATED** 5% glyphosate, HT
				VT Double PRO® corn: STRUCTURED 5% corn belt / 20% EPA cotton counties
	Trecepta® corn	MON89034 (Cry1A.105 + Cry2Ab), MIR162 (Vip3Aa)	glyphosate (NK603)	**Trecepta RIB Complete corn: INTEGRATED** 5% glyphosate, HT
				Trecepta corn: STRUCTURED 5% corn belt / 20% EPA cotton counties
	Agrisure® 3120 E-Z Refuge® corn	TC1507 (Cry1F), Bt11 (Cry1Ab)	glyphosate (GA21), glufosinate (pat)	**INTEGRATED** 5% glyphosate, glufosinate HT
	Agrisure® 3330 E-Z Refuge® corn	Bt11 (Cry1Ab), MIR162 (Vip3Aa) MON89034 (Cry1A.105 + Cry2Ab)	glyphosate (GA21), glufosinate (pat)	**INTEGRATED** 5% glyphosate, glufosinate HT

† PowerCore® Ultra Enlist® trait and PowerCore® Ultra Enlist® Refuge Advanced® trait will not be offered for sale or distribution until completion of field testing and applicable regulatory reviews.

* All integrated refuge products in the Pioneer® brand portfolio are available in EPA cotton counties with an additional 20% structured refuge.

ABOVE- AND BELOW-GROUND INSECT PROTECTION

		Above-ground insect protection event (protein)	Below-ground insect protection event (protein/gene)	Herbicide tolerance (HT) (event/gene)	Refuge requirement - Corn belt unless otherwise designated
Options in the Pioneer® brand portfolio	**Qrome® PRODUCTS**	DP4114 (Cry1F), MON810 (Cry1Ab)	DP4114 (Cry34/35Ab1), MIR604 (mCry3A)	glyphosate (NK603), glufosinate (pat)	**INTEGRATED** 5% glyphosate, glufosinate HT
	Optimum® AcreMax® XTreme PRODUCTS	TC1507 (Cry1F), MON810 (Cry1Ab)	DAS59122 (Cry34/35Ab1), MIR604 (mCry3A)	glyphosate (NK603), glufosinate (pat)	**INTEGRATED** 5% glyphosate, glufosinate HT
	Vorceed™ Enlist® corn PRODUCTS	DP4114 (Cry1F), MON89034 (Cry1A.105 + Cry2Ab)	DP4114 (Cry34/35Ab1), MON87411 (DvSnf7 + Cry3Bb1)	2,4-D choline + FOP (DAS40278), glyphosate (MON87411), glufosinate (pat)	**INTEGRATED** 5% 2,4-D choline, glyphosate, glufosinate, FOP HT
Competitive products	**SmartStax® Technology**	TC1507 (Cry1F), MON89034 (Cry1A.105 + Cry2Ab)	MON88017 (Cry3Bb1), DAS59122 (Cry34/35Ab1)	glyphosate (MON88017), glufosinate (pat)	**INTEGRATED** 5% glyphosate, glufosinate HT
	SmartStax® PRO Technology	TC1507 (Cry1F), MON89034 (Cry1A.105 + Cry2Ab)	MON87411 (DvSnf7 + Cry3Bb1), DAS59122 (Cry34/35Ab1)	glyphosate (MON87411), glufosinate (pat)	**INTEGRATED** 5% glyphosate, glufosinate HT
	VT4PRO™ with RNAi Technology	MON89034 (Cry1A.105 + Cry2Ab2) and MIR162 (Vip3Aa20)	MON87411 (DvSnf7 + Cry3Bb1)	glyphosate (MON88017)	**INTEGRATED** 5% glyphosate, glufosinate HT
	Agrisure® Duracade® 5122 E-Z Refuge® corn	TC1507 (Cry1F), Bt11 (Cry1Ab)	MIR604 (mCry3A), 5307 (eCry3.1Ab)	glyphosate (GA21), glufosinate (pat)	**INTEGRATED** 5% glyphosate HT

Efficacy levels based on Corteva Agriscience and/or independent university entomologist results against susceptible insect populations. Product responses can vary by location, pest population, environmental conditions, and agricultural practices.

PowerCore® multi-event technology developed by Corteva Agriscience and Monsanto. PowerCore® is a registered trademark of Monsanto Technology LLC. Always follow IRM, grain marketing and all other stewardship practices and pesticide label directions. B.t. products may not yet be registered in all states. Check with your seed representative for the registration status in your state.

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Liberty®, LibertyLink® and the Water Droplet Design are trademarks of BASF.

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Please contact your Pioneer sales professional for information and suggestions specific to your operation. Individual results may vary.

Following burndown, Enlist Duo® and Enlist One® herbicides with Colex-D® technology are the only herbicides containing 2,4-D that are authorized for preemergence and postemergence use with Enlist® crops. Consult Enlist® herbicide labels for weed species controlled. Enlist Duo and Enlist One herbicides are not registered for use or sale in all states and counties; are not registered in AK, CA, CT, HI, ID, MA, ME, MT, NH, NV, OR, RI, UT, VT, WA and WY; and have additional subcounty restrictions in AL, GA, TN and TX, while existing county restrictions still remain in FL. All users must check "Bulletins Live! Two" no earlier than six months before using Enlist One or Enlist Duo. To obtain "Bulletins," consult epa.gov/espp/, call 1-844-447-3813, or email ESPP@epa.gov. You must use the "Bulletin" valid for the month and state and county in which Enlist One or Enlist Duo are being applied. Contact your state pesticide regulatory agency if you have questions about the registration status of Enlist® herbicides in your area. ALWAYS READ AND FOLLOW PESTICIDE LABEL DIRECTIONS. IT IS A VIOLATION OF FEDERAL AND STATE LAW TO USE ANY PESTICIDE PRODUCT OTHER THAN IN ACCORDANCE WITH ITS LABELING. ONLY USE FORMULATIONS THAT ARE SPECIFICALLY LABELED FOR SUCH USE IN THE STATE OF APPLICATION. USE OF PESTICIDE PRODUCTS, INCLUDING, WITHOUT LIMITATION, 2,4-D-CONTAINING PRODUCTS NOT AUTHORIZED FOR USE WITH ENLIST CROPS, MAY RESULT IN OFF-TARGET DAMAGE TO SENSITIVE CROPS/AREAS AND/OR SUSCEPTIBLE PLANTS, IN ADDITION TO CIVIL AND/OR CRIMINAL PENALTIES. Additional product-specific stewardship requirements for Enlist crops, including the Enlist Product Use Guide, can be found at www.traitstewardship.com.

Pioneer® brand products are provided subject to the terms and conditions of purchase which are part of the labeling and purchase documents.

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